

Pedigree Chart Inheritance Predictor

GitHub Repository:

<https://github.com/rutvikbarbhai/Pedigree-Chart-Inheritance-Predictor-2025>

1. Introduction

This project implements a machine learning model powered by a Convolutional Neural Network (CNN) to classify pedigree charts into one of six genetic inheritance patterns:

- Autosomal Dominant
- Autosomal Recessive
- X-Linked Dominant
- X-Linked Recessive
- Y-Linked Dominant
- Y-Linked Recessive

Pedigree charts graphically represent the inheritance of traits across generations. Automatically interpreting the chart and predicting its inheritance type provides value in genetics research and automated diagnostics. This model processes images of pedigree charts and predicts their inheritance type using CNNs.

2. GitHub Repository Link

The complete project, including source code, datasets, model files, and documentation, is available on GitHub at: <https://github.com/rutvikbarbhai/Pedigree-Chart-Inheritance-Predictor-2025>

3. Summary of CNN Implementation Steps

1. Dataset Preparation

A dataset of pedigree chart images was curated and augmented to include variations in rotation, scaling, and noise. The images were normalized and resized to a fixed input dimension (e.g., 224×224 pixels) to ensure consistency during training.

2. Data Pipeline

Utility scripts were developed to:

- Load and preprocess image data.
- Shuffle and batch data.
- Split data into training, validation, and testing subsets.

3. Model Design

The CNN was built using TensorFlow and Keras. The architecture includes:

- Convolutional layers with ReLU activation.
- Max-pooling layers to reduce spatial dimensions.
- Dropout layers for regularization.
- Fully connected dense layers with softmax output for six classes.

Hyperparameters such as learning rate, batch size, and optimizer (Adam) were tuned for optimal results.

4. Training and Validation

The model was trained using categorical cross-entropy loss and validated periodically. Checkpoints were saved for the best-performing models. Metrics like accuracy and loss were plotted for analysis.

5. Testing and Evaluation

The trained CNN was evaluated on a hold-out test set. A confusion matrix and classification report were used to measure precision, recall, and overall accuracy.

6. Deployment and Usage

A Python script enables inference on new pedigree chart images. The trained model predicts the likely inheritance type and displays the corresponding class label.

4. Insights, Challenges, and Solutions

Key Insights

- The CNN effectively distinguished between major inheritance patterns.
- Data augmentation significantly reduced overfitting and improved generalization.
- Some underrepresented classes (e.g., Y-linked patterns) required further data balancing.

Challenges and Resolutions

- **Class Imbalance:** Addressed using oversampling, augmentation, and class weighting.
- **Variability in Chart Styles:** Standardized image preprocessing and augmentation.
- **Overfitting:** Controlled using dropout, early stopping, and regularization techniques.
- **Interpretability:** Used confusion matrices and visual inspection to analyze misclassifications.

Future Improvements

- Expand dataset for inheritance types.
- Apply transfer learning using pretrained CNNs such as ResNet or EfficientNet.
- Integrate explainability tools like Grad-CAM for model transparency.
- Develop a mobile or web interface for real-time predictions.

5. Conclusion

This project demonstrates how convolutional neural networks can accurately predict inheritance patterns from pedigree chart images. It presents a complete workflow—from data collection to model deployment—documented in the GitHub repository. While the model performs well, further work in data expansion and explainability can enhance accuracy and usability. The modular architecture allows easy integration into clinical or educational genetic tools.